

KRITIK SOMAN

AI Specialist with 5+ years of experience | Canadian Permanent Resident

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in kritik-soman

PROFESSIONAL EXPERIENCE

AI/ML Computational Science Specialist - Accenture

📅 April 2022 – Present

📍 Bengaluru, India

- **ApprovalX:** Intelligent Compliance Automation Platform
 - Designed and deployed an agentic QA framework using **LangGraph** and **GPT-4o** for automated document compliance checks across legal, medical, and branding domains.
 - Spearheaded development of a scalable PoC for multi-channel compliance (email, banners), integrating 50+ detectors (e.g., font, brand color, logo, claims). Achieved production-grade throughput of 100+ requests/hour.
 - Engineered a multi-lingual font verification pipeline for e-commerce banners using HiSAM, Google OCR, and glyph matching—accurate down to 10px font size and low-contrast text.
- **InstructGen:** GenAI-Powered Marketing Automation Suite
 - Built a Retrieval-Augmented Generation (RAG) pipeline for brand-specific email **HTML generation** with custom CSS.
 - Developed a RAG-based **SQL generation** engine for customer targeting by merging sub-queries indexed in a vector database.
- **Xtracto:** Cross-Format Document Parsing Engine
 - Architected a standardized parser for diverse digital formats (PPTX, PSD, PDF, HTML, raster images), integrating tools like pdfminer, psd-tools, Aspose API, Playwright, pdf2image, Google OCR, Microsoft Florence VLM, and HiSAM.
 - Enabled LLM-based automation across heterogeneous document types, unlocking scalable insights and compliance validation.
- **Leadership & Mentorship:**
 - Lead a team of 4 AI Analysts through full-cycle development, driving iterative improvements and production readiness.
 - Mentored undergraduate and postgraduate research projects from IIT Patna and BITS Pilani, guiding thesis work in GenAI applications.
 - Conducted hiring and technical interviews for entry-level AI Analyst roles, contributing to team growth and talent acquisition.

Junior Data Scientist - App Orchid

📅 August 2020 – April 2022

📍 Hyderabad, India

- **NLP:** Created methods for extraction of custom entities in contracts based on Question Answering (QA), RegEx, Shallow Syntax Parsing, Token REGEX, Relative Time Expression, Language Identification, Fine tuning QA, Sentence similarity models, Classification of Document, Clauses.
- **CV:** Row and column detection in table images using cascade RCNN. Heading, text, table detection in contract images using YOLO v3+.

Intern - STMicroelectronics

📅 February 2016 – June 2016

📍 Greater Noida, India

- Undergraduate thesis on **Motion Algorithms using MEMS Sensors:** Worked on magnetometer calibration, robust heading estimation by fusing gyroscope and magnetometer data for Pedestrian Deduced Reckoning (PDR).

PATENTS/PUBLICATIONS

- Method and System for Modifying Images (US Patent Filed)
- Artificial Intelligence based System and Method for Document Analysis (US Patent Filed)
- Soman, Kritik. "Rooftop Detection using Aerial Drone Imagery." In *Proceedings of the ACM India Joint International Conference on Data Science and Management of Data*, pp. 281-284. 2019.

EDUCATION

M.Tech. in *Machine Learning*

Indian Institute of Technology, Kanpur

📅 August 2018 – June 2020 | GPA: 9.3/10

- Thesis on **Semantic Mobile Base Station Placement:** Developed a method using Genetic Algorithm and Semantic Segmentation for placement of mobile base stations at optimal location given RGB and depth aerial images. (Supervisor: Prof. K.S. Venkatesh)

B.Tech. in *ECE* - IIIT Noida

📅 August 2012 – June 2016 | GPA: 9.0/10

AWARDS

Accenture Technology Wizard Award 2025 | Excellent Mentor 2023 | Client Value Creation Award 2023 | IIT Kanpur Academic Excellence Award for 2018-19 | ACM CoDS-COMAD Travel Grant 2019, 2020

OPEN SOURCE



GIMP-ML 📄

Created GIMP-ML - Deep Learning Models for GNU Image Manipulation Program (1.5k ★ on GitHub), News: Visual Informatics Group @ UT Austin, Papers With Code (Top Trending), Presented at PyCon India 2024 @Google



VoiceFlow

A voice powered GUI automation tool using agents (part of Berkeley Advanced LLM MOOC - Ninja Tier)

FRAMEWORKS

PyTorch | Transformers | Spacy | NLTK | OpenCV | sklearn | skimage | Pandas | playwright | psd tools | sentence-transformers | Google Agent Development Kit | LangGraph | Postgres | Django | FastAPI | pdfminer | Git